

SOFTWARE ENGINEERING

LAB-1

PROBLEM STATEMENT-15

NAME: Ananya

CLASS: AIML Sec 'H' 5th Semester

SRN:PES1UG24AM474

PROBLEM STATEMENT- Pharmacy Expiry & Re-order Dispatch Engine

1. Introduction & Scenario Review

Introduction

The **Pharmacy Expiry & Re-order Dispatch Engine** is an automated stock management system designed for hospital pharmacies. The system helps manage medicine inventory by tracking batch expiry dates, generating **First-Expired-First-Out (FEFO)** dispensing lists, and initiating purchase orders when medicine stock falls below the required safety threshold.

Scenario Review

Hospital pharmacies need to ensure that medicines are dispensed in the correct order and that sufficient stock is maintained. The system therefore focuses on three key activities:

- **Expiry Tracking:** Track the expiry dates of individual medicine batches.

- **FEFO Dispensing:** Prioritize the medicine batch with the nearest expiry date during dispensing.
- **Automated Re-ordering:** Generate and dispatch supplier purchase-order requests when stock falls below the safety threshold.

The primary actors identified in the scenario are the **Pharmacy Clerk** and **Inventory Supplier**.

The main objective of the system is to **reduce medicine wastage due to expiry and prevent stock shortages through automated inventory monitoring and re-ordering**.

2. Elicit & Draft Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.	High	Pass: The system selects the valid medicine batch with the nearest expiry date when multiple unexpired batches are available. Fail: The system selects a newer-expiring batch while an older unexpired batch exists.	Ensures medicines closest to expiry are dispensed first, reducing medicine wastage due to expiry
FR-002	Functional	The system shall record and track the expiry date of each medicine batch stored in the pharmacy inventory.	High	Pass: When a medicine batch is added, its expiry date is stored and can be retrieved from the inventory. Fail: A medicine batch can be stored without an expiry date or its expiry date cannot be retrieved.	Accurate batch expiry information is required to identify expired medicines and implement FEFO dispensing.
FR-003	Functional	The system shall monitor the available stock quantity of each medicine and identify when the quantity falls below its configured safety threshold.	High	Pass: When the available quantity becomes lower than the configured safety threshold, the system flags the medicine for re-order. Fail: The system does not flag the medicine after its quantity falls below the safety threshold.	Stock monitoring allows the system to identify medicines that need replenishment before they run out.
FR-004	Functional	The system shall automatically generate a purchase order request when a medicine's stock quantity falls below its configured safety threshold.	High	Pass: A purchase order request is generated automatically when the stock quantity falls below the configured threshold.	Automated purchase-order generation reduces manual effort and helps prevent

				Fail: No purchase order request is generated after the stock falls below the configured threshold.	medicine shortages.
FR-005	Functional	The system shall dispatch generated purchase order requests to the appropriate inventory supplier.	Medium	Pass: A generated purchase order is sent to the configured inventory supplier and its dispatch status is recorded. Fail: The purchase order is generated but is not sent to the supplier or its dispatch status is not recorded.	Dispatching the purchase order ensures that the supplier receives the replenishment request.
NFR-001	Non-Functional	The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below the configured safety threshold.	High	Pass: Benchmarking tests confirm that purchase-order generation and dispatch meet the defined response-time target and security requirements under simulated peak load. Fail: The system exceeds the defined response-time target or fails the required security checks during simulated peak load.	Ensures timely replenishment while maintaining secure system operation during high workload conditions.
NFR-002	Non-Functional	The system shall maintain an audit record of medicine stock changes, batch expiry updates, and purchase order activities.	Medium	Pass: Every stock update, expiry-date update, and purchase-order activity creates a timestamped audit record containing the	Provides traceability of inventory and re-order activities and helps identify discrepancies

				performed action and its status. Fail: Any completed inventory or purchase-order activity occurs without a corresponding audit record.	in pharmacy records.
--	--	--	--	---	----------------------

3. Extract Actors & Use Cases:

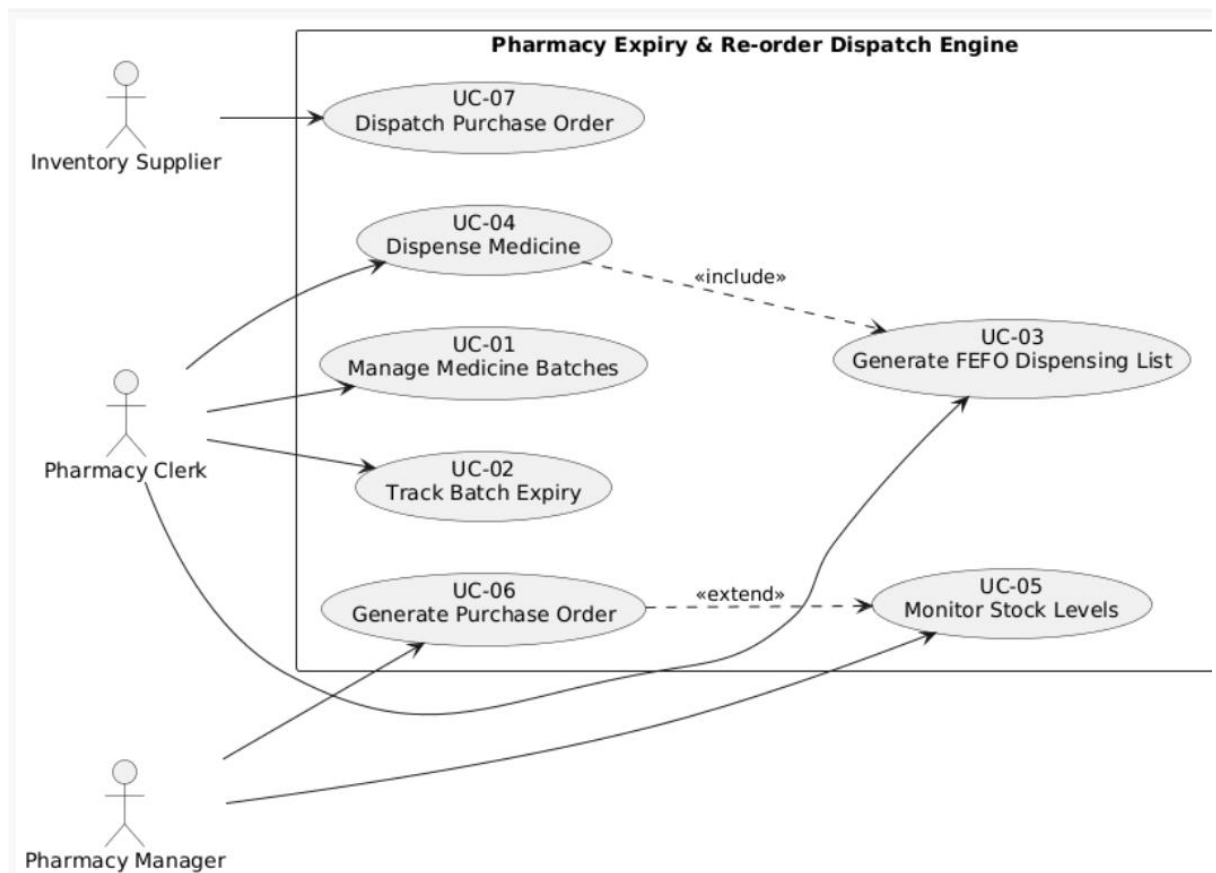
Actors

1. Pharmacy Clerk
2. Inventory Supplier
3. Pharmacy Manager

Use cases

Use Case ID	Use Case	Related Requirement
UC-01	Manage Medicine Batches	FR-002
UC-02	Track Batch Expiry	FR-002
UC-03	Generate FEFO Dispensing List	FR-001
UC-04	Dispense Medicine	FR-001
UC-05	Monitor Stock Levels	FR-003
UC-06	Generate Purchase Order	FR-004
UC-07	Dispatch Purchase Order	FR-005

4. Draw at least one UML Use-Case Diagram:



5. Write One Use-Case Flow:

Use Case: UC-04 – Dispense Medicine

Preconditions

- The Pharmacy Clerk is authorized to dispense medicine.
- The requested medicine is available in the pharmacy inventory.
- At least one unexpired medicine batch is available.
- Batch expiry information is recorded in the system.

Postconditions

- The required medicine quantity is deducted from the selected batch.

- The inventory quantity is updated.
- The dispensing transaction is recorded successfully.

Main Success Scenario

1. Pharmacy Clerk selects the medicine to be dispensed.
2. The system retrieves all available batches of the selected medicine.
3. The system checks the expiry dates of the available batches.
4. The system identifies the valid batch with the nearest expiry date according to FEFO.
5. The system displays the selected batch to the Pharmacy Clerk.
6. Pharmacy Clerk confirms the required quantity.
7. The system deducts the dispensed quantity from the selected batch.
8. The system updates the medicine inventory.
9. The system records the dispensing transaction.
10. The system displays a confirmation that the medicine has been successfully dispensed.

Alternate Flow – No Valid Batch Available

1. The system determines that all available batches of the requested medicine are expired.
2. The system prevents the dispensing operation.
3. The system displays a message indicating that no valid batch is available.
4. Pharmacy Clerk selects another medicine or ends the dispensing operation.